

EC-467 Mobile Application Development



BS (CE)-2021 LAB REPORT #9

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

Lab Instructor.

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LAB REPORT #9

View Pager

Introduction:

View Pager is a class/layout manager that allows the user to swipe/flip left and right through pages of application.

Adapter:

The adapter is used to generate the fragment or pages on the View Pager.

Tabs:

To make the View Pager more user friendly, the tabs can be added on top or bottom of the view pager.

Tools:

Android Studio Jellyfish

Objectives of the experiment:

- 1.To get basic understanding regarding view pager.
- 2.To get basic understanding regarding how fragments used in view pager.
- 3.To get to know about the benefits related to view pager.

Class Task

JAVA CODE OF Main Activity:

```
package com.huzaifashahbaz.viewpagerapp;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.viewpager.widget.ViewPager;
import com.google.android.material.tabs.TabLayout;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

```

        ViewPagerAdapter adapter=new
ViewPagerAdpter(getSupportFragmentManager());
        ViewPager viewPager=findViewById(R.id.view_pager);
        TabLayout tabLayout=findViewById(R.id.tabs);
        tabLayout.setupWithViewPager(viewPager);
        viewPager.setAdapter(adapter);
    }
}

```

XML CODE OF MAIN ACTIVITY:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <com.google.android.material.tabs.TabLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/tabs"/>
    <androidx.viewpager.widget.ViewPager
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@+id/view_pager"/>
</LinearLayout>

```

JAVA CODE OF FRAGMENT A:

```

package com.huzaifashahbaz.viewpagerapp;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
import java.nio.Buffer;
public class FragmentA extends Fragment {
    @Nullable
    @Override

```

```

    public View onCreateView(@NonNull LayoutInflater inflater, @Nullable
    ViewGroup container, @Nullable Bundle savedInstanceState)
    {
        View
        fragmentView=inflater.inflate(R.layout.fragment_a_layout,container,false);
        return fragmentView;
    }
}

```

XML CODE OF FRAGMENT A:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="@dimen/view_margin"
    android:background="@color/green">
    <TextView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@+id/fragment_title_a"
        android:gravity="center"
        android:textSize="32sp"
        android:textColor="@color/green"
        android:text="Fragment A"/>
</LinearLayout>

```

JAVA CODE OF FRAGMENT B:

```

package com.huzaifashahbaz.viewpagerapp;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
public class FragmentB extends Fragment {
    @Nullable
    @Override

```

```

    public View onCreateView(@NonNull LayoutInflater inflater, @Nullable
    ViewGroup container, @Nullable Bundle savedInstanceState)
    {
        View
        fragmentView=inflater.inflate(R.layout.fragment_b_layout,container,false);
        return fragmentView;
    }
}

```

XML CODE OF FRAGMENT B:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="@dimen/view_margin"
    android:background="@color/blue">
    <TextView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@+id/fragment_title_b"
        android:gravity="center"
        android:textSize="32sp"
        android:textColor="@color/white"
        android:text="FragmentB"/>
</LinearLayout>

```

JAVA CODE OF VIEWPAGERADAPTER:

```

package com.huzaifashahbaz.viewpagerapp;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentPagerAdapter;
public class ViewPagerAdapter extends FragmentPagerAdapter{
    public ViewPagerAdapter(FragmentManager fm)
    {
        super(fm);
    }
}

```

```

@NonNull
@Override
public Fragment getItem(int position)
{
    if (position==0)
    {
        return new FragmentA();
    }
    else if(position==1)
    {
        return new FragmentB();
    }
    else
    {
        return null;
    }
}

@Override
public int getCount()
{
    return 2;
}

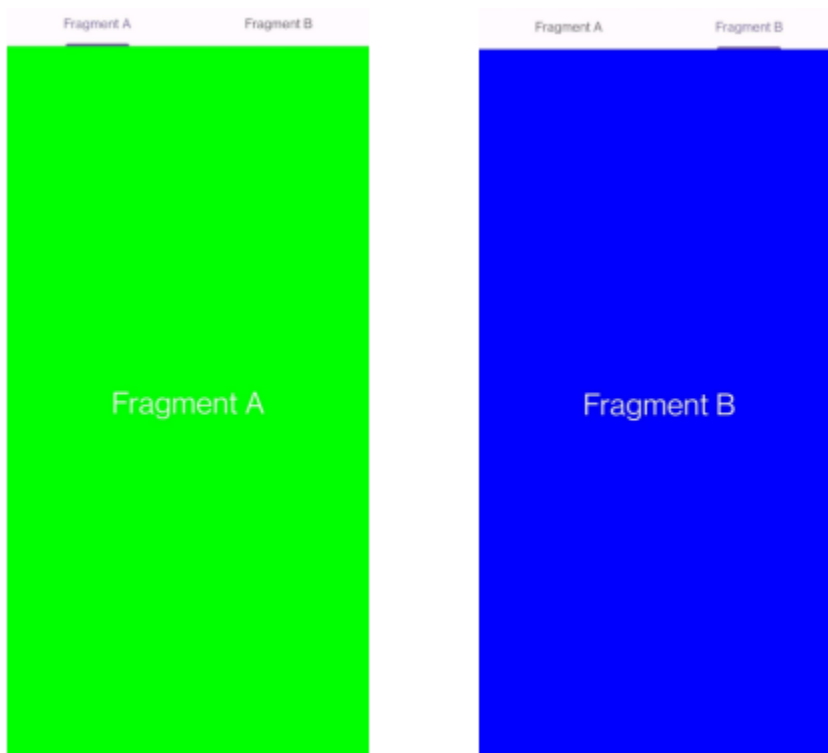
@Nullable
@Override
public CharSequence getPageTitle(int position)
{
    if(position==0)
    {
        return "FragmnetA";
    }
    else if (position==1)
    {
        return "FragmentB";
    }
    else
    {
        return null;
    }
}
}

```

Results:



OUTPUT:



Home Task:

Use Tasbeeh app and show history in separate fragment in view pager.

Note: Bundle can share string array list.

JAVA CODE OF Main Activity:

```
package TASBBEHVIEWPAGER;  
import android.os.Bundle;  
import androidx.annotation.NonNull;  
import androidx.appcompat.app.AppCompatActivity;  
import androidx.fragment.app.Fragment;
```

```

import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentPagerAdapter;
import androidx.viewpager.widget.ViewPager;
import com.google.android.material.tabs.TabLayout;
import com.huzaifashahbaz.viewpagerapp.R;

import java.util.ArrayList;
import java.util.List;
public class MainActivity extends AppCompatActivity {
    private ViewPager viewPager;
    private TabLayout tabLayout;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        viewPager = findViewById(R.id.view_pager);
        tabLayout = findViewById(R.id.tab_layout);
        setupViewPager(viewPager);
        tabLayout.setupWithViewPager(viewPager);
    }
    private void setupViewPager(ViewPager viewPager) {
        ViewPagerAdapter adapter = new
ViewPagerAdapter(getSupportFragmentManager());
        adapter.addFragment(new CounterFragment(), "Counter");
        adapter.addFragment(new HistoryFragment(), "History");
        viewPager.setAdapter(adapter);
    }
    class ViewPagerAdapter extends FragmentPagerAdapter {
        private final List<Fragment> mFragmentList = new ArrayList<>();
        private final List<String> mFragmentTitleList = new ArrayList<>();

        public ViewPagerAdapter(@NonNull FragmentManager fm) {
            super(fm);
        }

        @NonNull
        @Override
        public Fragment getItem(int position) {
            return mFragmentList.get(position);
        }
    }

```



```

@Override
public int getCount() {
    return mFragmentManager.size();
}

public void addFragment(Fragment fragment, String title) {
    mFragmentManager.add(fragment);
    mFragmentManager.add(title);
}

@Override
@Override
public CharSequence getPageTitle(int position) {
    return mFragmentManager.get(position);
}
}
}

```

XML CODE OF Main Activity:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <com.google.android.material.appbar.AppBarLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:theme="@style/ThemeOverlay.AppCompat.Dark.ActionBar">
        <com.google.android.material.tabs.TabLayout
            android:id="@+id/tab_layout"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            app:tabIndicatorColor="@android:color/white"
            app:tabSelectedTextColor="@android:color/white"
            app:tabTextColor="@android:color/darker_gray" />
        </com.google.android.material.appbar.AppBarLayout>
    <androidx.viewpager.widget.ViewPager

```

```

        android:id="@+id/view_pager"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1" />
</LinearLayout>

```

JAVA CODE OF HISTORYFRAGMENT:

```

package TASBBEHVIEWPAGER;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
import com.huzaifashahbaz.viewpagerapp.R;
public class HistoryFragment extends Fragment {
    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater, @Nullable
        ViewGroup container, @Nullable Bundle savedInstanceState) {
        View view = inflater.inflate(R.layout.activity_fragment_history, container,
        false);
        Toast.makeText(getContext(), "History will be displayed here",
        Toast.LENGTH_SHORT).show();
        return view;
    }
}

```

XML CODE OF HISTORYFRAGMENT:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:padding="16dp">
</LinearLayout>

```

JAVA CODE OF COUNTERFRAGMENT:

```
package TASBBEHVIEWPAGER;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
public class CounterFragment extends Fragment {
    private TextView tvCount;
    private int count = 0;
    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater, @Nullable
    ViewGroup container, @Nullable Bundle savedInstanceState) {
        View view = inflater.inflate(R.layout.fragment_counter, container, false);
        tvCount = view.findViewById(R.id.text_view_count);
        Button btnCount = view.findViewById(R.id.button_count);
        Button btnReset = view.findViewById(R.id.button_reset);
        btnCount.setOnClickListener(v -> {
            count++;
            tvCount.setText("Count: " + count);
        });
        btnReset.setOnClickListener(v -> {
            count = 0;
            tvCount.setText("Count: " + count);
        });
        return view;
    }
}
```

XML CODE OF COUNTERFRAGMENT:

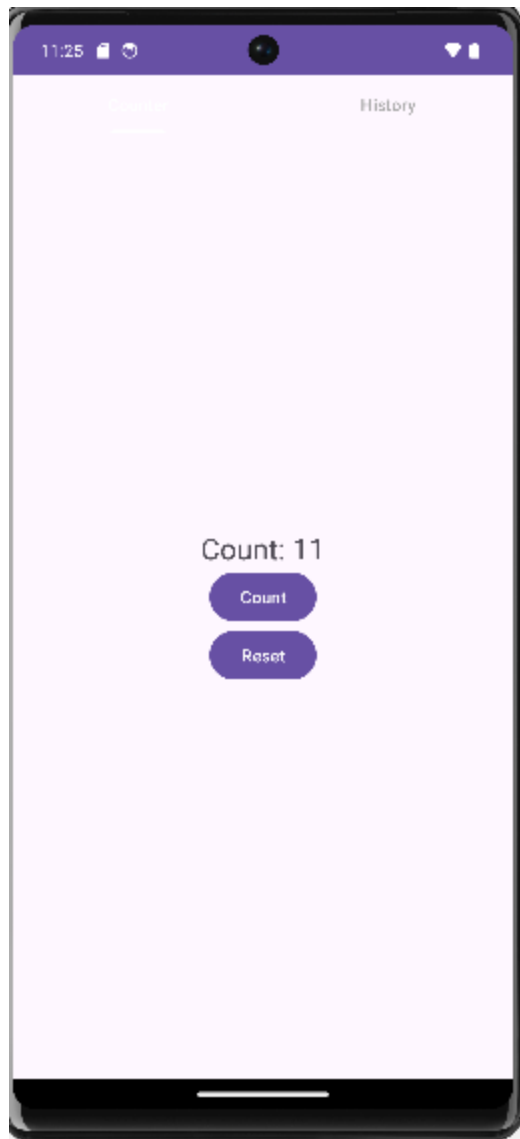
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

```
        android:gravity="center"
        android:padding="16dp">
        <TextView
            android:id="@+id/text_view_count"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Count: 0"
            android:textSize="24sp" />
        <Button
            android:id="@+id/button_count"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Count" />
        <Button
            android:id="@+id/button_reset"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Reset" />
    </LinearLayout>
```

Results:



OUTPUT:



Conclusion:

In conclusion, we learn how to make a viewpager app and also I made a tasbeeh viewpager by using a counter and history fragment by using java code and xml linear layout.